15.Write a Pandas program to keep the rows with at least 2 NaN values in a given DataFrame.

AIM:

To write a Pandas program to keep the rows that contain at least 2 missing (NaN) values in a given DataFrame.

DATASET:

Name	Age	Marks	City	Grade
Anu	20	85	Chennai	A
Bala	NaN	NaN	Madurai	B
Cathy	21	NaN	NaN	A
David	NaN	78	NaN	B
Esha	20	88	Chennai	A

Python code:

```
import pandas as pd
import numpy as np
data = {
    "Name": ["Anu", "Bala", "Cathy", "David", "Esha"],
    "Age": [20, np.nan, 21, np.nan, 20],
    "Marks": [85, np.nan, np.nan, 78, 88],
    "City": ["Chennai", "Madurai", np.nan, np.nan, "Chennai"],
    "Grade": ["A", "B", "A", "B", "A"]
}
df = pd.DataFrame(data)
print("Original DataFrame:")
print(df)
# Keep rows having at least 2 NaN values
result = df[df.isnull().sum(axis=1) >= 2]
print("\nRows with at least 2 NaN values:")
print(result)
```

Output:

```
Rows with at least 2 NaN values:
    Name   Age  Marks     City Grade
1   Bala   NaN    NaN  Madurai     B
2  Cathy  21.0    NaN      NaN     A
3  David   NaN   78.0      NaN     B
```

Result:

Rows containing at least two missing values were successfully extracted.

16.Write a Pandas program to split the following dataframe into groups based on school code. Also check the type of GroupBy object.

Aim:

A Pandas program to split a DataFrame into groups based on school code and check the type of the GroupBy object.

Dataset:

School_Code	Class	Student	Age
S001	10A	Anu	15
S001	10B	Bala	16
S002	10A	Cathy	15
S002	10B	David	16
S003	10A	Esha	15
S003	10B	Farhan	16

Python code:

```python
import pandas as pd
data = {
    "School_Code": ["S001", "S001", "S002", "S002", "S003", "S003"],
    "Class": ["10A", "10B", "10A", "10B", "10A", "10B"],
    "Student": ["Anu", "Bala", "Cathy", "David", "Esha", "Farhan"],
    "Age": [15, 16, 15, 16, 15, 16]
}
df = pd.DataFrame(data)
print("Original DataFrame:")
print(df)
# Group by School_Code
grouped = df.groupby("School_Code")
print("\nType of GroupBy object:")
print(type(grouped))
print("\nGroups:")
for school, group in grouped:
    print("\nSchool Code:", school)
    print(group)
```

Output:

```
Groups:

School Code: S001
  School_Code Class Student  Age
```

```
School Code: S001
  School_Code Class Student  Age
0        S001   10A     Anu   15
1        S001   10B    Bala   16
```

```
School Code: S002
  School_Code Class Student  Age
2        S002   10A   Cathy   15
3        S002   10B   David   16
```

```
School Code: S003
  School_Code Class Student  Age
4        S003   10A    Esha   15
5        S003   10B  Farhan   16
```

Result:

The DataFrame was successfully divided into groups based on school code, and the resulting object was identified as a DataFrameGroupBy object.